Smart Incentive Spirometer - carnatic music

A.Karthik(2023102009) K.Roshan Lal(2023102010) P.Deepak(2023101109)

Aim

To develop a sensor-based smart incentive spirometer that detects and displays the user's breathing effort as different ball levels on a web application, provides level-based audio and Carnatic music feedback to motivate the user during repetitive lung exercises, and stores the exercise data to generate a personalized dashboard for tracking individual progress over time.

Motivation

Lung exercises such as incentive spirometry are often **repetitive**, which can make the activity monotonous and make it difficult for users to remain motivated and practice consistently.

Conventional incentive spirometers mainly provide **immediate visual feedback** through mechanical ball movement, but the repetitive nature of daily exercises can make them boring and difficult to stay motivated with, while the user's exercise performance and progress remain **untracked** over time.





Our motivation is to make the exercise more **interactive, measurable, and engaging** by sensing the user's breathing effort and displaying the corresponding ball level on a web application. When the user reaches different pressure levels, the system will provide **level-based audio feedback and Carnatic music** as a motivational reward, encouraging the user to put in greater effort and continue practicing.

The collected exercise data will be stored in a database and used to create a **personalized dashboard**, allowing users to view their performance, compare sessions, and track their progress over time.

Methodology

The proposed system will use a **blow pipe, pressure sensor, Arduino, web application, and database** to monitor and motivate the user's breathing exercise.

1. **Breathing Exercise:** The user blows through the blow pipe, generating pressure that is measured by the pressure sensor.
2. **Pressure Sensing:** The pressure sensor continuously measures the pressure generated by the user and sends the readings to the Arduino.
3. **Arduino Processing:** The Arduino collects and processes the sensor readings and sends the measured pressure data to the web application.
4. **Virtual Ball Display:** The web application will represent the user's pressure as the upward movement of a virtual ball. As the user generates higher pressure, the ball will move higher on the screen and cross predefined levels. Each level will correspond to a specific pressure threshold.
5. **Level-Based Audio Feedback:** If the user does not reach the minimum pressure threshold, the application will play a **noise/encouragement sound**. Once the user crosses the defined thresholds, **Carnatic music corresponding to the achieved level** will be played to provide positive feedback and motivate the user to increase their effort.
6. **Data Storage:** Exercise data such as pressure, achieved level, time spent, maximum pressure, and maximum-level hold duration will be stored in a database for each user.
7. **Progress Dashboard:** The stored data will be used to create a personalized dashboard showing the user's performance and progress across different exercise sessions.

Technical Implementation

Components

Hardware	Software
Blow pipe	Web Application
Pressure sensor	Backend/API
Aurdino	Database
Connecting wires	Audio/Music Module
Breadboard	Progress Dashboard

Tech Stack

Component	Technology
Microcontroller	Arduino
Frontend	React,Typescript
Backend	Python (FastAPI/Flask)
Database	PostgreSQL / MySQL
Dashboard	Web-based charts and visualizations
Audio	Browser-based audio playback

Login & User Tracking

Each user will have an individual account so that their exercise sessions can be stored separately and their **personalized progress dashboard** can be generated.

The dashboard will track parameters such as:

- Time spent
- Maximum pressure achieved
- Maximum level reached
- Hold duration at maximum level
- Number of repetitions/sessions
- Daily and overall progress

Systematic Review &Research

a. Lung Capacity & Respiratory Exercise

- Understand lung capacity, inspiratory volume, airflow, and pressure.
- Study how incentive spirometers and breathing exercises work.
- Understand what our pressure measurement represents and avoid assuming it directly measures lung capacity.

b. Motivation for Repetitive Lung Exercises

- Study why users lose motivation when performing repetitive exercises.
- Research methods such as immediate feedback, goals, progress tracking, gamification, and rewards that can improve exercise adherence.

c. Music and Repetitive Activities

- Study how music affects motivation, perceived effort, mood, and performance during repetitive physical activities. Existing reviews generally report positive effects of music on motivation and exercise experience. Investigate whether **self-selected/preferred music** improves engagement compared with no music.

d. Carnatic Music as Feedback

- Research on Carnatic music can be effectively used as a **motivational/reward mechanism** in repetitive exercises.

e.Sensor & Device Research

Determine appropriate thresholds for the virtual ball levels.

Future Scope

- **Mobile App:** Develop a dedicated mobile app to track user progress and exercise history.
- **Bluetooth/Wi-Fi:** Connect the physical device to the mobile app using Bluetooth or Wi-Fi.
- **Standalone Device:** Develop a physical device that can measure the user's effort and play music directly.
- **Music in Device:** Integrate a speaker into the device to play level-based Carnatic music without depending on the web app.
- **New Product:** Develop the complete system as a standalone smart incentive spirometer product.
- **Scalability:** Extend the system to support multiple users and devices with individual progress tracking.

Project Timeline

- **Week 1-2:** Research and finalize project requirements, device design, sensors, and music-feedback approach.
- **Week 3-4:** Build the blow-pipe and Arduino prototype with the pressure sensor.
- **Week 5:** Calibrate the sensor, define pressure levels, and establish Arduino-to-web communication.
- **Week 6-7:** Develop the web application with login, virtual ball visualization, and level-based audio/Carnatic music feedback.
- **Week 8:** Implement database integration and personalized progress dashboard.
- **Week 9-10:** Evaluate the prototype, make improvements, and complete documentation.

References:

- Motivation & Music Interventions in Adults
- Effects of Music Choice on Performance and Psychophysiological Responses to Exercise
- https://publications.ersnet.org/content/erj/60/1/2101499

- Music during anaerobic exercise in physically active adults: task-dependent evidence for repetition performance

Example

Suppose there are **3 levels** :

- **Level 0:** Minimum threshold not reached → Noise/encouragement
- **Level 1:** First threshold reached →Carnatic music 1
- **Level 2:** Second threshold reached →Carnatic music 2
- **Level 3:** Highest threshold reached →Carnatic music 3

The pressure levels/thresholds and the corresponding Carnatic music nodes/levels are currently under discussion with Saroja mam and will be finalized by next sprint.